Written exam 17 December, 2020

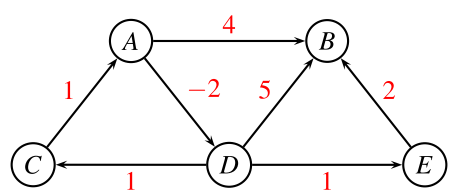
Course title: Network Optimization **Course no:** 42115

Aids allowed: All
Exam duration: 4 hours
Weighting: The points are indicated at each question
Language: You can write in Danish or English

Individual exam: You are not allowed to work together or contact any other people. All solutions that are suspiciously similar, will be reported to the study administration.

Shortest path problem

Consider the following directed graph where distances w_{ij} are marked with red. You are asked to find the shortest path from A to all other nodes.



Problem 1: (1 point) Argue whether the problem contains negative cycles, and state which algorithm you would use to solve the problem. ■

Problem 2: (3 point) Solve the problem. Give sufficient details to make it possible to follow the algorithm. ■

Project planning

Consider a critical path problem, in which each task i has a processing time p_i, and a list of predecessors P_i. The predecessors P_i need to finish before task i can begin.

task	predecessor(s)	processing time p_i
A	C	8
B	C,E	5
C		4
D	A,B,F	3
E		2
F	B,H	4
G	A	2
H	E	6

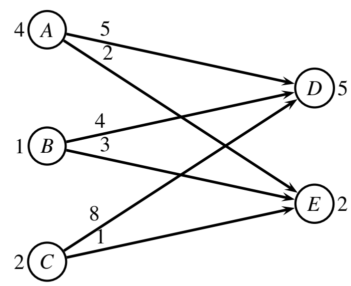
Problem 3: (2 point) Order the activities in topological order. ■

Problem 4: (3 point) For each activity, find the earliest starting time and the latest starting time. Also, report the makespan T. Show the recursion you are using. ■

Problem 5: (1 point) Mark the critical path, and argue why the edges are on the critical path. ■

Transportation problem

Three factories A, B, C are transporting items to two customers D, E. The *transportation costs* of sending one item along an edge is indicated on the graph below. The production at factories are respectively $p_A = 4, p_B = 1, p_C = 2$, while the demands at the customers are $d_D = 5, d_E = 2$.



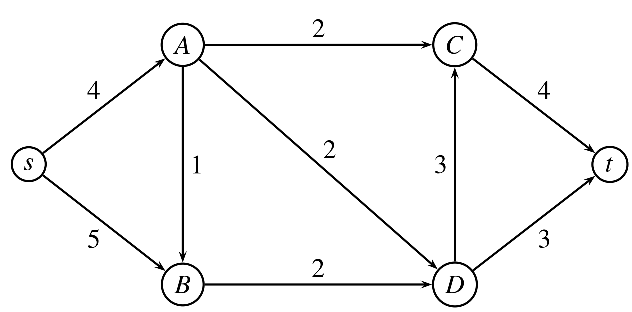
A greedy algorithm considers the factories in the order A, B, C, in each step sending the items to the customers in the cheapest possible way.

Problem 6: (2 point) Find the greedy solution, and report the cost of the solution. ■

Problem 7: (4 point) Use the network simplex algorithm to solve the problem to optimality. In each iteration report the dual variables, the reduced costs of edges. Report the cost of the optimal solution. ■

Maximum flow problem

In the following network the edge weights denote the capacity. The problem is to find the maximum flow from s to t.



Problem 8: You can answer *one* of the following questions:

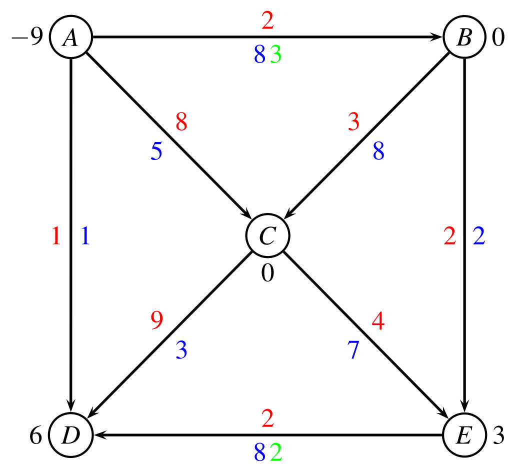
- (2 points) Use the augmenting path algorithm to solve the above problem.
- (4 points) Use the preflow-push (also called push-relabel) algorithm to solve the above problem.

In both cases, give sufficient details to make it possible to follow the algorithm. ■

Problem 9: (1 point) Describe how you find a minimum cut, and report the cut, and its capacity. ■

Repositioning of containers

Consider the following network with edge costs c_{ij} (red) and edge capacities u_{ij} (blue). Moreover, we have lower bounds ℓ_{ij} (green) on some of the edges. The demand b_i (black) in every node is positive for a sink, and negative for a source.



(∗)

Problem 10: (2 point) Transform the min-cost flow problem with upper and lower bounds to an ordinary min-cost flow problem, having only upper bounds. Describe the individual steps of the transformation. ■

Now, consider the transformed problem. Having solved a maximum flow problem to find a feasible solution we are told that $T = \{(A,B),(B,C),(C,E),(E,D)\}$ and $U = \{(A,D),(B,E),(C,D)\}$ and $L = \{(A,C)\}$. Edges in T form a tree. The flow on edges in U is at the upper bound, the flow on edges in L is at the lower bound (i.e. zero).

Problem 11: (2 point) Calculate the flow corresponding to the feasible solution defined by T,U,L. ■

Problem 12: You can answer *one* of the following questions:

- (3 point) Use the augmenting cycle algorithm to solve the problem to optimality.
- (4 point) Use the network simplex algorithm to solve the problem to optimality. In each iteration report the dual variables, the reduced costs of edges.

In each iteration report the selected cycle, and how much can be flown along the cycle. Report the cost of the optimal solution. ■

Problem 13: (2 point) Find the solution to the original problem (*). Report the flow on the edges, and the total cost. ■

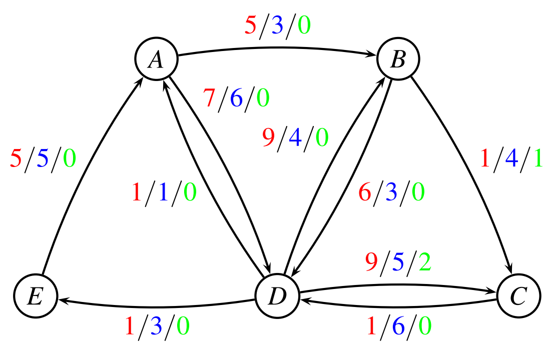
Flowing containers in a liner shipping network

The below instance, BALTICLINE, shows a small liner shipping network. Nodes represent ports, and edges represent legs sailed by vessels. The capacities indicate the remaining capacity of the vessels after some mandatory orders have been fulfilled. Therefore, the capacity may vary for each leg.

Note: The network is the same as in *Project Assignment 2020*, but the (source,terminal) pairs for each commodity is changed.

In mathematical terms, we are given a weighted oriented graph $G = (V, E, c, u, \ell)$ where each edge $(i, j) \in E$ has an associated cost (per container) $c_{ij} \geq 0$, a capacity $u_{ij} \geq 0$, and a lower bound $\ell_{ij} \geq 0$.

BALTICLINE **(Route net of a Baltic liner shipping company)**



For edge (i, j) the first number (red) is the cost c_{ij}, the second number (blue) is the capacity u_{ij}, and the third number (green) is the lower bound ℓ_{ij}.

We have the demands

k	s_k	t_k	d_k
1	D	C	2
2	D	B	1
3	A	E	3

Commodity $k = 1$ is shipping two containers from D to C, commodity $k = 2$ is shipping one container from D to B, and commodity $k = 3$ is shipping three containers from A to E. The commodities should be transported *unsplittable*, i.e. all d_k units should follow the same path.

Problem 14: (2 point) For each commodity k in BALTICLINE, write up a table with all possible simple routes from s_k to t_k through the graph $G = (V, E, c, u, \ell)$. A route may only use edges where the capacity is at least d_k, and it needs to be simple (i.e. cycle-free). For each commodity k and each feasible route state the corresponding cost of sending the d_k containers. (**Hint:** There should be 9 distinct routes.) ■

Problem 15: (3 point) Write up the path formulation of BALTICLINE. ■

Now, assume that we want to solve the LP-relaxed version of the path formulation with column generation. After a few iterations of the algorithm (using dummy routes as initial columns) we have the following dual solution:

Dual variables for the upper bound constraints: $y_{AB} = -2$, $y_{DA} = -3$. Dual variables for the lower bound constraints: $y'_{BC} = 7$, $y'_{DC} = 2$. Dual variables for the commodity constraints: $\overline{y}_1 = 8$, $\overline{y}_3 = 2$. All other dual variables are zero.

Problem 16: (3 point) Formulate the pricing problem of commodity $k = 1$ as a shortest-path problem. Draw the corresponding graph, indicate the costs of the edges, and mark the root r and the sink s. ■

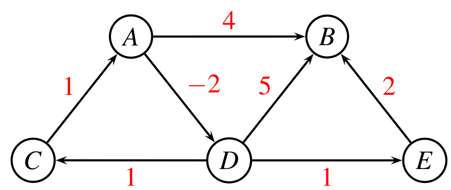
Organizing a conference

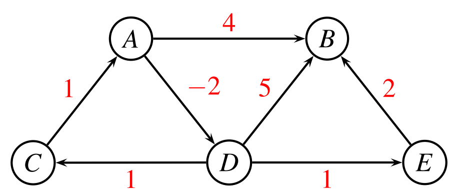
Suppose you are organizing a conference where researchers present articles they have written. Researchers who want to present an article send a paper to the conference organizers. The conference organizers have access to a committee of reviewers $R = \{r_1, \dots, r_m\}$. Each reviewer r_i is willing to read up to a_i articles. There are $P = \{p_1, \dots, p_n\}$ paper submissions, and each paper must be reviewed by 3 to 5 distinct reviewers (more reviewers is better). Moreover, each paper submission has a particular topic and each reviewer has a specialization for a set of topics, so papers on a given topic only get reviewed by those reviewers who are experts on that topic. The conference organizers need to decide which reviewers will review each article.

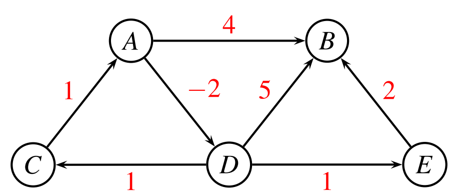
Problem 17: (4 point) Explain how the assignment of reviewers to papers can be formulated as a network optimization problem. Describe the network (nodes, edges, edge weights, demands) and state which type of problem it is. ■

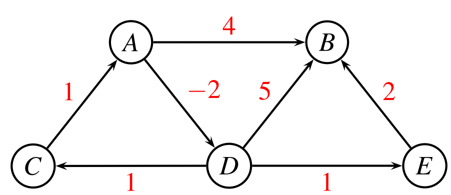
end of assignment

Shortest path problem

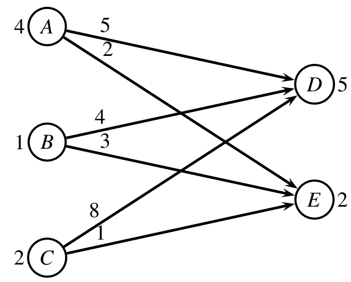


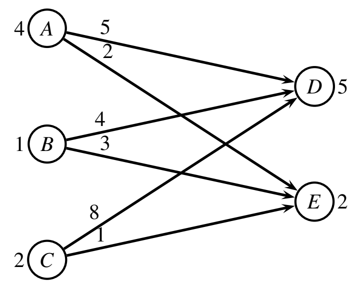


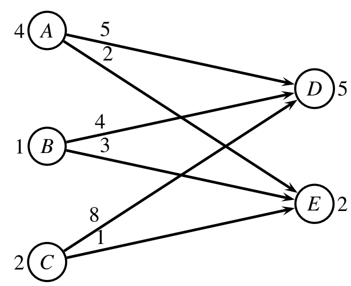


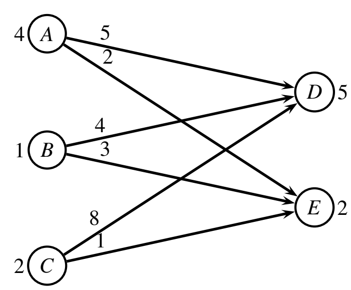


Transportation problem

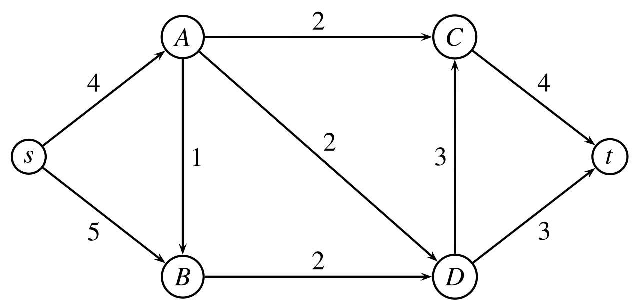


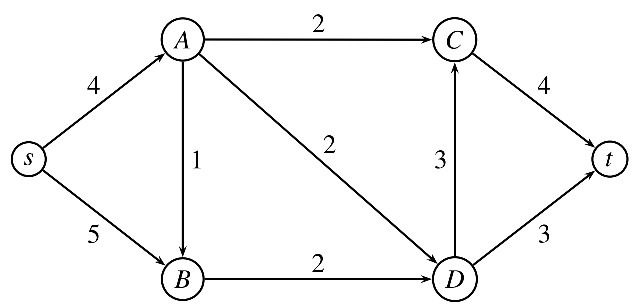


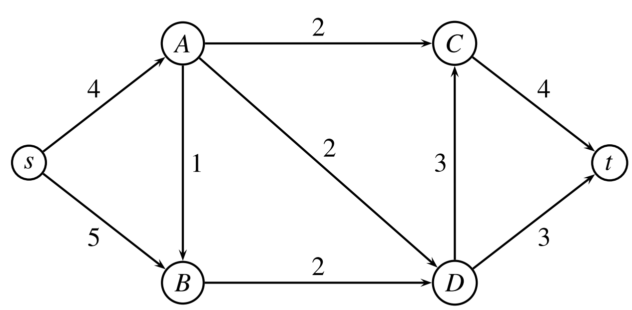


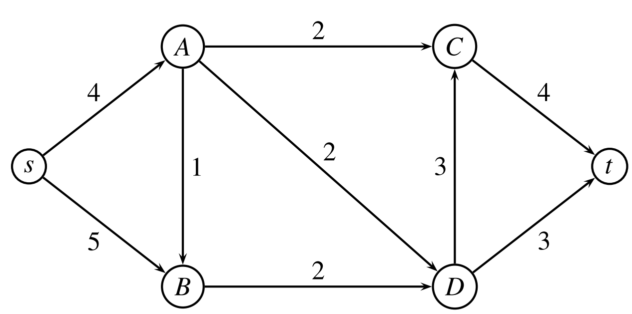


Maximum flow problem









Repositioning of containers